

An Exploratory Analysis of Limit Order Book Data

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ABSTRACT

In quantitative finance, forecasting equity prices from microstructure data remains a fundamental open problem. This paper presents a systematic time- and frequency-domain analysis of Limit Order Book (LOB) data for Intel Corporation (INTC, November 3, 2014) obtained via the LOBSTER platform, followed by the estimation and comparison of three parametric models: an autoregressive process $AR(p)$, an ordinary-least-squares multivariable linear regression, and a vector autoregression $VAR(p)$. Model selection is performed via the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Empirically, the $VAR(1)$ model attains the best information-criterion scores among those considered. A key finding is that partitioning the trading day by realized-volatility regime—separating the opening half-hour from the mid-session—materially reduces in-sample criterion values and residual autocorrelation, suggesting that intraday non-stationarity is a first-order modeling concern for LOB data.

1 DATA DESCRIPTION

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be an underlying probability space and let $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ be a filtration representing the information generated by the exchange up to time t . The LOB is a priority-queued matching mechanism in which resting limit orders are aggregated at discrete price levels. LOBSTER snapshots the state of the book every $\Delta t = 100$ ms, yielding an evenly-spaced discrete time series $\{X_k\}_{k=0}^N$ where $t_k = k \Delta t$.

DEFINITION 1 (LEVEL- ℓ MIDPRICE). Let $b_\ell(t)$ and $a_\ell(t)$ denote the best bid and ask prices at depth level $\ell \in \{1, \dots, 5\}$ at time t . The level- ℓ midprice is

$$M_\ell(t) = \frac{b_\ell(t) + a_\ell(t)}{2}.$$

Unless stated otherwise, $M(t) \equiv M_1(t)$ refers to the level-1 (best-quote) midprice.

DEFINITION 2 (SIGNED MARKET ORDER FLOW). Let $I_k \in \{+1, -1\}$ be the trade-direction indicator at event k (buy = +1, sell = -1) and let $V_k^{MO} > 0$ be the corresponding executed volume (number of shares). The signed market-order flow at snapshot k is

$$\Phi_k = I_k \cdot V_k^{MO} \in \mathbb{Z} \setminus \{0\}.$$

The primary datasets recorded at each 100 ms snapshot k are summarized in Table 1.

2 HYPOTHESES

- H1. Models targeting $\Delta P_t^{(\ell)} := M_\ell(t) - M_\ell(t - \Delta t)$ at level $\ell > 1$ will yield strictly larger AIC and BIC than comparable models targeting $\Delta P_t^{(1)}$.

- H2. Coarsening the observation interval to $\delta t \gg \Delta t$ will increase the statistical significance of estimated coefficients, because price drift accumulates over longer horizons.
- H3. Atheoretic time-series models (AR, VAR) will attain lower in-sample AIC and BIC than ordinary least-squares (OLS) regression models, owing to the martingale-like character of the price process, which violates the exogeneity assumptions underlying OLS.
- H4. Within the OLS framework, signed market-order flow Φ_t will be a more significant regressor for ΔP_t than total volume $V_t = V_t^B + V_t^A$, because Φ_t encodes directional information lost in the unsigned aggregate.

3 DATA ANALYSIS

All time series are indexed by snapshot index $k \in \{0, 1, \dots, N\}$ with $t_k = k \cdot 100$ ms. Full time-series plots appear in the Appendix.

3.1 Autocorrelation Structure

Let $\{Z_k\}$ be a zero-mean weakly stationary process. Its *autocovariance function* (ACVF) and *autocorrelation function* (ACF) are defined, respectively, as

$$\gamma_Z(h) = \text{Cov}(Z_{k+h}, Z_k), \quad \rho_Z(h) = \frac{\gamma_Z(h)}{\gamma_Z(0)}, \quad h \in \mathbb{Z}.$$

The *partial autocorrelation function* (PACF) at lag h , denoted $\pi_Z(h)$, is the last coefficient of the best linear predictor of Z_{k+h} given $(Z_{k+1}, \dots, Z_{k+h-1})$:

$$\pi_Z(h) = (Z_{k+h} - \hat{Z}_{k+h}, Z_k - \hat{Z}_k),$$

where $\hat{Z}_{k+h}$ and $\hat{Z}_k$ denote the respective projections onto $\text{sp}\{Z_{k+1}, \dots, Z_{k+h-1}\}$.

Signed market-order flow Φ_k . The sample ACF of $\{\Phi_k\}$ decays monotonically toward zero by lag ~ 10 , consistent with a low-order AR process. The PACF exhibits statistically significant spikes at

Table 1: LOB Snapshot Fields (100 ms cadence, INTC, 2014-11-03)

Field	Type	Description
NumberMO	$\mathbb{Z}_{\geq 0}$	Count of market orders in window
VolumeMO	$\mathbb{Z}_{\geq 0}$	Total shares executed by market orders
EventTime	$\mathbb{R}_{>0}$	Milliseconds since midnight
BuyPrice	$\mathbb{R}^5$	Bid prices at levels 1–5
SellPrice	$\mathbb{R}^5$	Ask prices at levels 1–5
BuyVolume	$\mathbb{Z}_{\geq 0}^5$	Bid depth at levels 1–5
SellVolume	$\mathbb{Z}_{\geq 0}^5$	Ask depth at levels 1–5
MO	list	Individual market-order records

lags 3 and 5, suggesting $\Phi_k \sim \text{AR}(p)$ with $p \in \{3, 5\}$ as leading candidates (Figure 2).

Differenced level-1 midprice ΔP_k . The sample ACF of $\{\Delta P_k\}$ is consistent with a white-noise process: no lag is individually significant at the 5% level except a marginal spike near lag 11 (Figure 3). The PACF confirms this—no significant partial autocorrelations appear below lag 11—suggesting that $\{\Delta P_k\}$ is at minimum close to an uncorrelated sequence, motivating examination of an $\text{AR}(p)$ with $p \leq 11$.

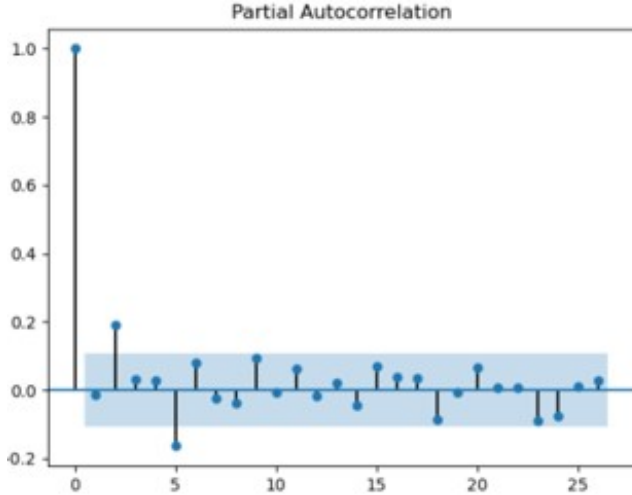


Figure 1: Sample PACF of signed market-order flow Φ_k . Dashed bands indicate pointwise 95% confidence intervals under the null $H_0 : \pi(h) = 0$ computed as $\pm 1.96/\sqrt{N}$.

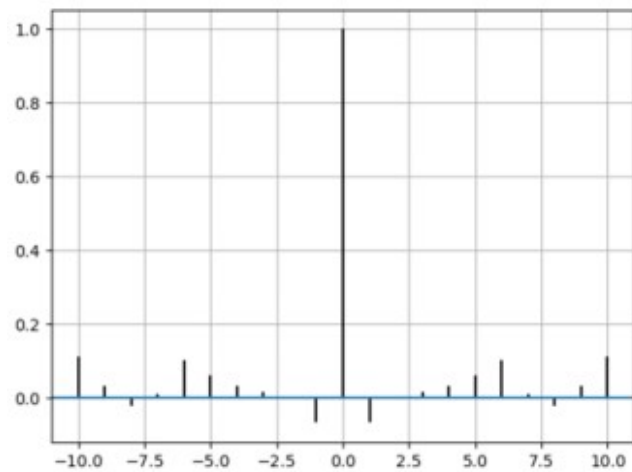


Figure 2: Sample ACF of ΔP_k (differenced level-1 midprice). The process is consistent with white noise for all lags except a marginal exceedance near lag 11.

3.2 Cross-Correlation Analysis

The cross-correlation function between processes $\{X_k\}$ and $\{Y_k\}$ is

$$\rho_{XY}(h) = \frac{\text{Cov}(X_{k+h}, Y_k)}{\sqrt{\text{Var}(X_k) \text{Var}(Y_k)}}, \quad h \in \mathbb{Z}.$$

Note that $\rho_{XY}(h) \neq \rho_{YX}(h)$ in general; the sign of h distinguishes predictive from reactive relationships.

Φ_k and ΔP_k . The cross-correlation $\rho_{\Phi, \Delta P}(h)$ is significant at negative lags $h < 0$, indicating that lagged values Φ_{k+h} (with $h < 0$, i.e., past market-order flow) predict current price changes ΔP_k (Figure 4). Specifically, $\rho_{\Phi, \Delta P}(h) < 0$ for $h < 0$: an increase in Φ (net buy pressure) is associated with a subsequent *decrease* in ΔP , suggesting mean-reversion following aggressive buying.

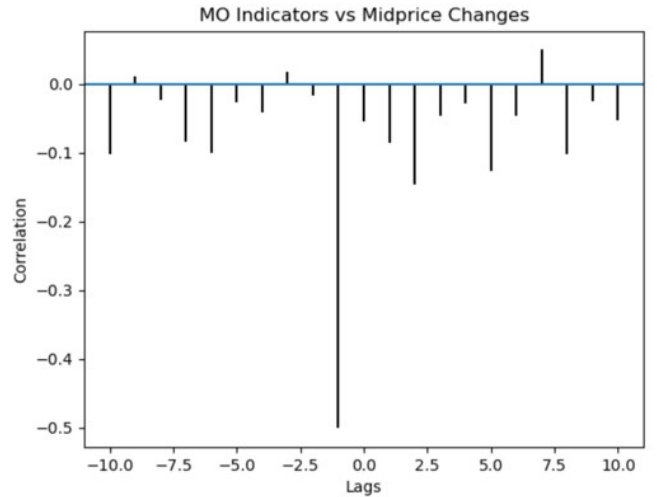


Figure 3: Sample cross-correlation $\hat{\rho}_{\Phi, \Delta P}(h)$. The significant negative values at $h < 0$ indicate that lagged signed order flow Granger-causes midprice change.

Φ_k and total volume V_k . Let $V_k = V_k^B + V_k^A$ denote total depth aggregated across the bid and ask. Because V_k exhibits a deterministic intraday trend, we partial it out by computing the detrended series $\tilde{V}_k = V_k - \hat{\mu}(t_k)$, where $\hat{\mu}$ is a nonparametric trend estimate. The cross-correlation $\rho_{\tilde{V}, \Phi}(h)$ shows that $\tilde{V}_{k+h}$ at $h > 0$ predicts Φ_k : total volume leads market-order direction. Moreover, $\rho_{\tilde{V}, \Phi}(h) < 0$, meaning that larger aggregate depth precedes a net reduction in buy-side market orders—counter to the naive intuition that volume and buy pressure should be positively co-determined (Figure 5).

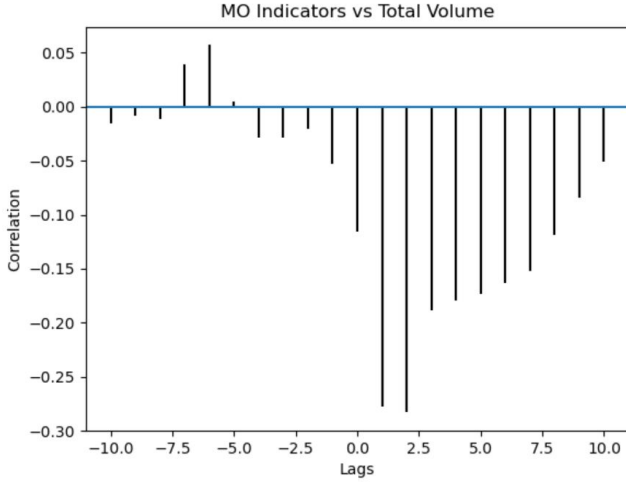


Figure 4: Sample cross-correlation $\hat{\rho}_{\tilde{V}, \Phi}(h)$. Significant negative values at positive lags indicate that detrended total volume Granger-causes signed market-order flow in the negative direction.

3.3 Spectral Analysis

The *spectral density* of a weakly stationary process $\{Z_k\}$ is the discrete-time Fourier transform of its ACVF:

$$f_Z(\lambda) = \sum_{h=-\infty}^{\infty} \gamma_Z(h) e^{-i\lambda h}, \quad \lambda \in [-\pi, \pi].$$

The *periodogram* is the finite-sample estimator $\hat{f}_Z(\lambda_j) = N^{-1} |d_Z(\lambda_j)|^2$, where $d_Z(\lambda_j) = \sum_{k=0}^{N-1} Z_k e^{-i\lambda_j k}$ is the discrete Fourier transform evaluated at the Fourier frequency $\lambda_j = 2\pi j/N$. Smoothed variants replace the raw periodogram by a lag-window estimator $\tilde{f}_Z(\lambda) = \sum_{|h| \leq M} \kappa(h/M) \hat{\gamma}_Z(h) e^{-i\lambda h}$, where κ is a data taper (e.g., Bartlett, Hanning, Boxcar).

Applying this to ask volume V_k^A , the raw periodogram concentrates nearly all power near frequency zero ($\lambda \approx 0$), even after mean removal. This indicates a slowly varying, trend-like component that is not fully eliminated by simple demeaning. The residual low-frequency power is attributable to heteroskedastic intraday volatility: the opening batch auction at 09:30 and the pre-close period near 16:00 generate systematic variance clusters that manifest as long-memory-like spectral mass near zero (Figure 6). This motivates the volatility-regime partitioning described in Section 4.

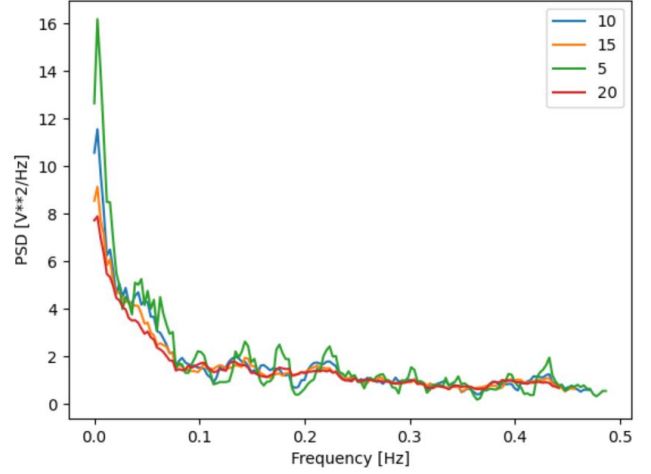


Figure 5: Boxcar-windowed power spectral density of $V_k^A - \bar{V}^A$. The persistent peak near $\lambda = 0$ indicates residual low-frequency energy after mean removal, consistent with intra-day nonstationarity.

4 MODELING

Information criteria. All models are compared via the Akaike Information Criterion

$$\text{AIC} = -2\hat{\ell} + 2q$$

and the Bayesian Information Criterion

$$\text{BIC} = -2\hat{\ell} + q \log N,$$

where $\hat{\ell}$ is the maximized log-likelihood and q is the number of free parameters. Lower values indicate a better bias-variance trade-off. BIC imposes a heavier penalty on q for $N > 7$ and is therefore consistent for model order selection under standard regularity conditions.

4.1 Autoregressive Model AR(p)

DEFINITION 3 (AR(p) PROCESS). $\{\Delta P_k\}$ follows an AR(p) process if there exist constants $\alpha_1, \dots, \alpha_p \in \mathbb{R}$ and white noise $\{\varepsilon_k\} \sim \text{WN}(0, \sigma^2)$ such that

$$\Delta P_k = \sum_{j=1}^p \alpha_j \Delta P_{k-j} + \varepsilon_k.$$

The process is causal (and thus weakly stationary) if and only if all roots of the characteristic polynomial $\phi(z) = 1 - \alpha_1 z - \dots - \alpha_p z^p$ lie strictly outside the closed unit disk: $\phi(z) = 0 \Rightarrow |z| > 1$.

Parameters are estimated via Yule-Walker equations, which equate sample autocovariances to their theoretical counterparts:

$$\hat{\alpha} = \hat{\Gamma}_p^{-1} \hat{\gamma}_p,$$

where $\hat{\Gamma}_p = [\hat{\gamma}(i-j)]_{i,j=1}^p$ is the sample Toeplitz autocovariance matrix and $\hat{\gamma}_p = (\hat{\gamma}(1), \dots, \hat{\gamma}(p))^T$. The Yule-Walker estimator is guaranteed to yield a causal solution because $\hat{\Gamma}_p$ is positive definite for a non-degenerate sample.

Estimation results for $p \in \{1, 2, 3\}$ are reported in Table 2.

Table 2: Yule-Walker AR(p) model comparison for ΔP_k .

Model	AIC	BIC
AR(1)	-7.838	-7.815
AR(2)	-7.829	-7.795
AR(3)	-7.949	-7.903

The AR(1) model yields the best BIC, consistent with the near-white-noise PACF structure observed in Section 3.1, and is preferred on grounds of parsimony: the Levinson-Durbin recursion confirms that incremental reductions in $\hat{\sigma}^2$ are negligible beyond $p = 1$. The fitted model is

$$\widehat{\Delta P}_k = \hat{\alpha}_1 \Delta P_{k-1},$$

with $|\hat{\alpha}_1| \ll 1$, indicating near-unit-root behavior in the midprice level but stationary differences. In-sample predictions display systematically lower variance than the realized series, a consequence of the optimal linear predictor shrinking toward the unconditional mean under near-white-noise conditions (Figure 7).

REMARK 1 (OPENING-HOUR NONSTATIONARITY). *The in-sample residuals exhibit a pronounced variance spike during the first ~ 60 minutes of trading, consistent with the stylized fact that intraday volatility follows a U-shaped pattern (see [3]). This violates the homoskedasticity assumption embedded in Yule-Walker estimation and motivates the regime-partitioned VAR in Section 4.3.*

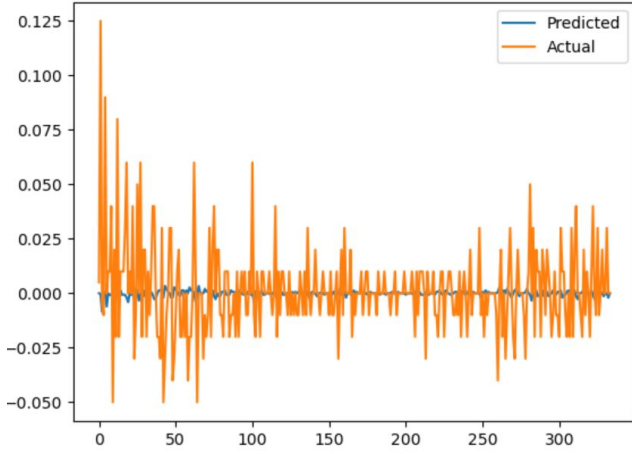


Figure 6: In-sample AR(1) fitted values $\widehat{\Delta P}_k$ (dashed) vs. realized ΔP_k (solid). The model under-estimates variance throughout, with pronounced mismatch during the market open.

4.2 Multivariable Linear Regression

DEFINITION 4 (OLS REGRESSION MODEL). Let $X_k \in \mathbb{R}^n$ be a vector of contemporaneous regressors observed at snapshot k , and suppose

$$\Delta P_k = \alpha^\top X_k + \varepsilon_k, \quad \varepsilon_k \sim \text{i.i.d. } \mathcal{N}(0, \sigma^2), \quad \mathbb{E}[\varepsilon_k | X_k] = 0.$$

The OLS estimator is $\hat{\alpha} = (X^\top X)^{-1} X^\top \Delta P$, which coincides with the MLE under the Gaussian error assumption.

Two specifications are evaluated:

$$(M1): \quad \Delta P_k = \alpha_1 V_k + \alpha_2 \Phi_k + \varepsilon_k, \quad (1)$$

$$(M2): \quad \Delta P_k = \alpha_1 \Phi_k + \varepsilon_k. \quad (2)$$

Both specifications return $\text{AIC} \approx -1677$ and $\text{BIC} \approx -1669$ —substantially worse than the AR models—confirming that contemporaneous regressors carry negligible explanatory power for ΔP_k once the near-martingale structure of the price process is accounted for. The large AIC/BIC values are consistent with a severely misspecified likelihood: the OLS assumption $\mathbb{E}[\varepsilon_k | X_k] = 0$ is violated when X_k and ΔP_k are determined simultaneously within the same limit-order book state.

Specifically, a necessary condition for OLS consistency is strict exogeneity of the regressors:

$$\mathbb{E}[\varepsilon_k | X_1, \dots, X_N] = 0 \quad \forall k,$$

which is unlikely to hold for V_k or Φ_k because both are contemporaneously determined alongside ΔP_k by the same order-arrival process. The residual plot (Figure 8) confirms this: fitted values cluster near zero for all k , and the residuals are virtually indistinguishable from the raw ΔP_k series.

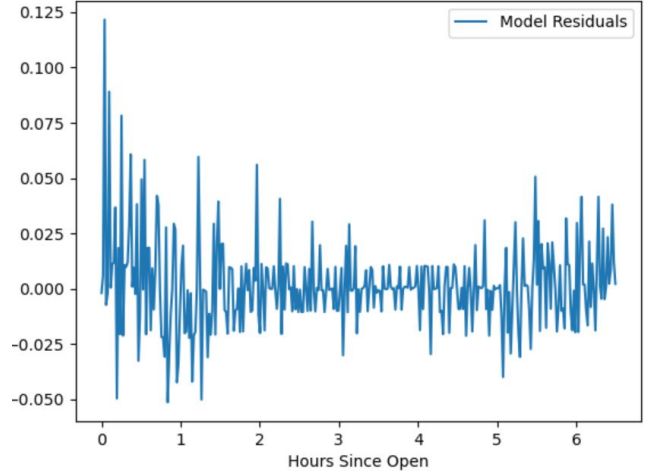


Figure 7: Residuals from the OLS model (M1). The near-zero fitted values indicate that neither V_k nor Φ_k explains contemporaneous midprice changes, consistent with the strict exogeneity violation discussed in the text.

4.3 Vector Autoregression VAR(p)

DEFINITION 5 (VAR(p) PROCESS). Let $y_k = (\Delta P_k, y_{1,k}, \dots, y_{n-1,k})^\top \in \mathbb{R}^n$. The process $\{y_k\}$ follows a VAR(p) if

$$y_k = c + \sum_{j=1}^p A_j y_{k-j} + \varepsilon_k,$$

where $c \in \mathbb{R}^n$ is a constant vector, $A_j \in \mathbb{R}^{n \times n}$ for $j = 1, \dots, p$, and the innovation sequence satisfies

$$\mathbb{E}[\varepsilon_k] = 0, \quad \mathbb{E}[\varepsilon_k \varepsilon_k^\top] = \Omega > 0, \quad \mathbb{E}[\varepsilon_k \varepsilon_{k-\ell}^\top] = 0 \quad \forall \ell \neq 0.$$

The process is stable (i.e., weakly stationary) if and only if all roots of $\det(I_n - \sum_{j=1}^p A_j z^j) = 0$ lie strictly outside the unit disk.

The VAR extends the AR by jointly modeling the endogenous interactions among components of $\mathbf{y}_k$. In particular, ΔP_k is allowed to be predicted by lagged values of all other endogenous variables, and vice versa. This makes the VAR appropriate when cross-equation Granger-causal relationships are present—as suggested by the cross-correlations in Section 3.2.

Identification and estimation. Parameters $(A_1, \dots, A_p, \Omega)$ are estimated by equation-by-equation OLS (equivalent to MLE under Gaussian innovations):

$$\hat{A} = \left(\sum_{k=p}^N \mathbf{y}_{k-1} \mathbf{y}_{k-1}^\top \right)^{-1} \sum_{k=p}^N \mathbf{y}_{k-1} \mathbf{y}_k^\top,$$

where $\mathbf{y}_{k-1} = (\mathbf{y}_{k-1}^\top, \dots, \mathbf{y}_{k-p}^\top)^\top$ is the stacked lag vector. The covariance estimator is $\hat{\Omega} = N^{-1} \sum_{k=p}^N \hat{\varepsilon}_k \hat{\varepsilon}_k^\top$.

Regime partitioning. Motivated by the spectral evidence in Section 3.3 and the AR residual analysis in Section 4.1, the trading day is divided into two regimes:

Regime I (mid-session): $t \in [1, 5.5]$ hours after market open. Total volume V_k is detrended and the terminal volume spike at $t \approx 5.5$ h is excluded. This yields the VAR(1):

$$\begin{bmatrix} \Delta P_k \\ \tilde{V}_k \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{bmatrix} \Delta P_{k-1} \\ \tilde{V}_{k-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,k} \\ \varepsilon_{2,k} \end{bmatrix}.$$

Regime II (market open): $t \in [0, 1]$ hour after open. Signed market-order flow Φ_k is used in place of V_k , as Φ_k requires no detrending over this short horizon:

$$\begin{bmatrix} \Delta P_k \\ \Phi_k \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{bmatrix} \Delta P_{k-1} \\ \Phi_{k-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,k} \\ \varepsilon_{2,k} \end{bmatrix}.$$

Results. For Regime I, MLE yields AIC = -2.88845 and BIC = -2.88803 . While the α_{12} coefficient associated with $\tilde{V}_{k-1}$ in the ΔP equation is not statistically significant at conventional levels, residual diagnostics show a clean white-noise pattern (Figure 9), confirming that the partitioning successfully eliminates systematic residual autocorrelation. For Regime II, residuals of the ΔP equation retain marginal autocorrelation at several lags (Figure 10), consistent with a richer AR structure during the high-volatility open; this suggests that a VAR(p) with $p > 1$ or an ARMAX specification may be warranted in that regime.

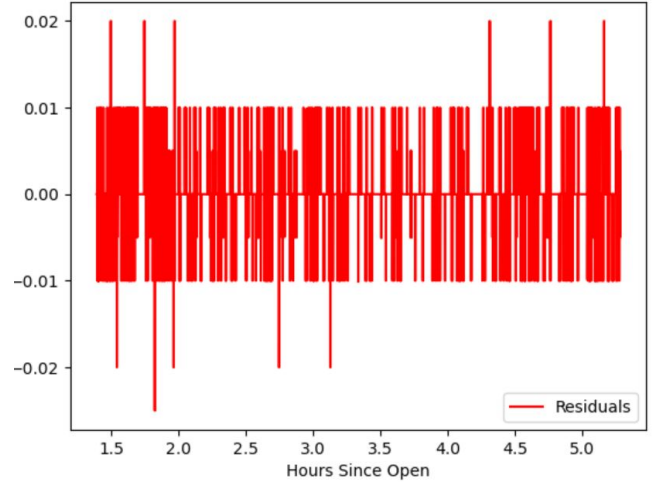


Figure 8: VAR(1) residuals for Regime I (mid-session, V_k as second endogenous variable). Sample ACF of $\hat{\varepsilon}_{1,k}$ is consistent with white noise at all plotted lags.

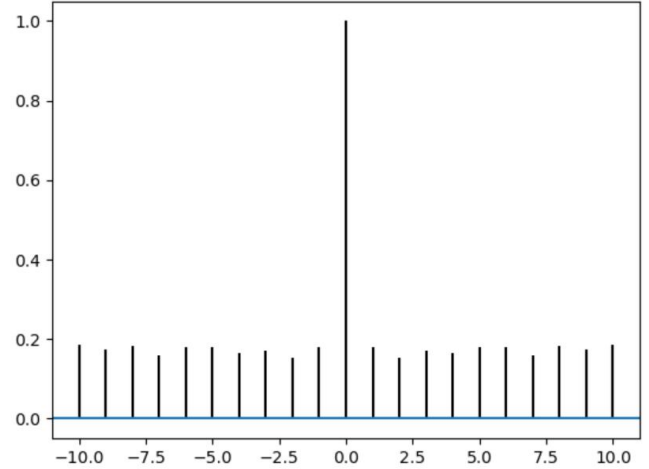


Figure 9: Sample ACF of residuals from the Regime II VAR(1) (Φ_k as second endogenous variable). Persistent autocorrelation at later lags indicates residual dynamics not fully captured by the VAR(1).

5 CONCLUSION

This analysis delivers several empirically grounded findings, most of which contradict the stated hypotheses.

- (1) **Against HH1.:** The order book level ℓ has negligible effect on model fit. AIC and BIC are essentially invariant to the choice $\ell \in \{1, \dots, 5\}$, likely because Intel is a highly liquid large-cap stock with negligible depth asymmetry across levels.
- (2) **Against HH2.:** Coarsening the time interval from $\Delta t = 100$ ms to $\delta t \approx 70$ s did not materially improve coefficient significance.

- (3) **In favor of HH3.:** The VAR(1) dominates the OLS regression on both AIC and BIC by a wide margin, consistent with the failure of strict exogeneity in the OLS specifications.
- (4) **Inconclusive on HH4.:** Neither Φ_k nor V_k is a significant contemporaneous predictor of ΔP_k in the OLS framework. However, lagged Φ_k does appear to Granger-cause ΔP_k , confirming the cross-correlation evidence in Section 3.2.

The most consequential finding, which was not hypothesized *a priori*, is that intraday volatility regime partitioning substantially improves model diagnostics. Excluding the opening hour reduces residual autocorrelation and improves information criteria for the VAR, corroborating the empirical literature on U-shaped intraday volatility [3]. This suggests that a two-regime model—one for the high-volatility open, one for the relatively stationary mid-session—is a natural specification for LOB microstructure forecasting.

5.1 Directions for Future Work

- (1) **Regime-switching models.** A hidden Markov model or threshold VAR could endogenize the volatility-regime boundaries rather than imposing them by calendar time.
- (2) **Cross-asset comparison.** Replicating the analysis on higher-volatility instruments (e.g., small-cap equities, futures) would assess the generalizability of the regime-partitioning finding.
- (3) **Higher-order LOB features.** Order-book imbalance, queue position, and cancellation rates are microstructure covariates with theoretical justification as price-change predictors; their inclusion in the VAR endogenous vector is a natural extension.
- (4) **Out-of-sample evaluation.** In-sample information criteria do not guarantee out-of-sample forecast accuracy. A rolling-window or walk-forward evaluation on multi-day data would provide a more rigorous assessment of practical forecasting utility.
- (5) **ARCH/GARCH extensions.** The variance clustering documented in the spectral analysis motivates modeling Ω_k as time-varying via a GARCH-in-mean or BEKK specification.

6 WORKS CITED

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[3] Webb, R.I. and Smith, D.G. (1994). The effect of market opening and closing on the volatility of Eurodollar futures prices. *Journal of Futures Markets*, 14(1):51–78. <https://doi.org/10.1002/fut.3990140106>

7 APPENDIX

7.1 Time Series Plots

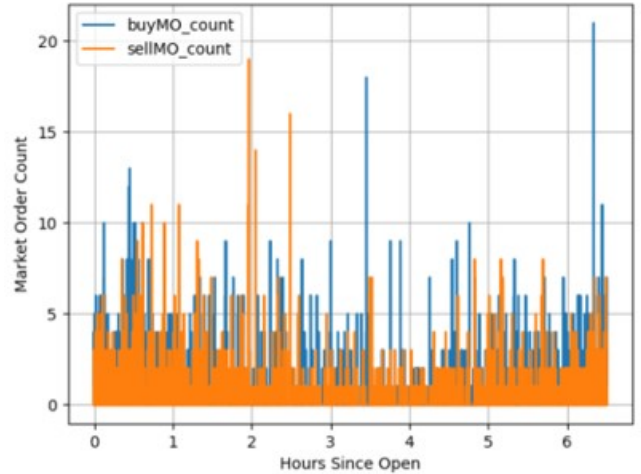


Figure 10: INTC NumberMO time series.

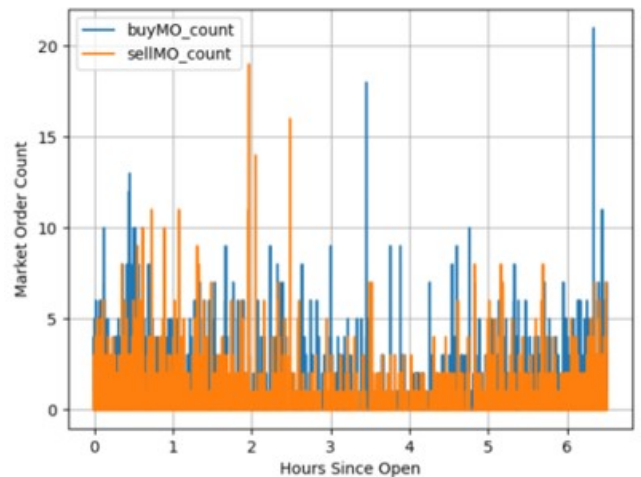


Figure 11: INTC VolumeMO time series.

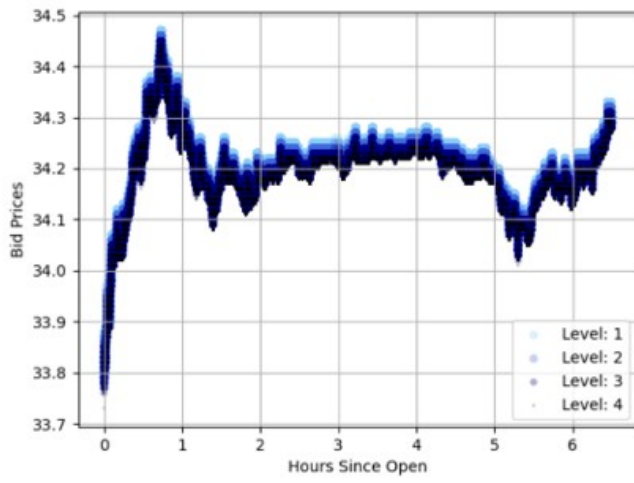


Figure 12: INTC bid prices $b_\ell(t)$, $\ell = 1, \dots, 5$.

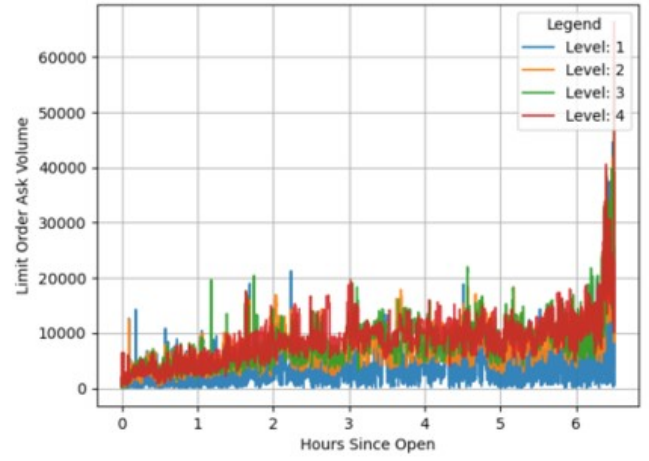


Figure 14: INTC bid depth V_k^B (level-aggregated).

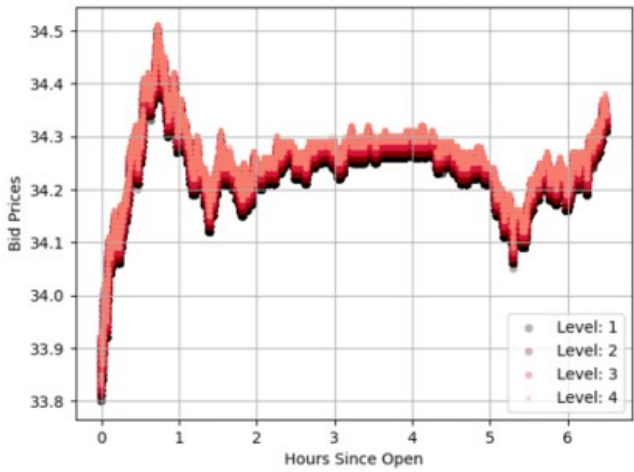


Figure 13: INTC ask prices $a_\ell(t)$, $\ell = 1, \dots, 5$.

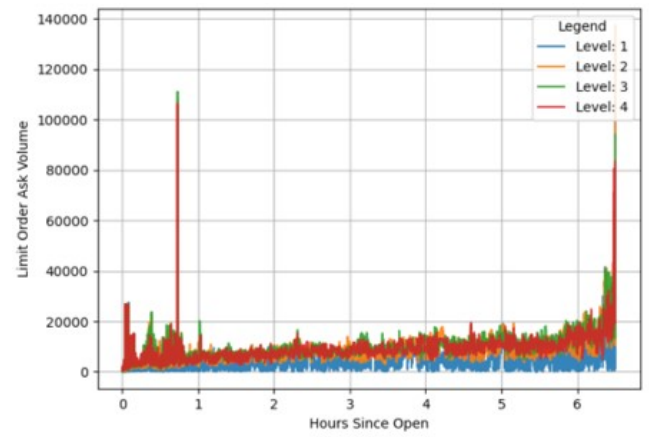


Figure 15: INTC ask depth V_k^A (level-aggregated).

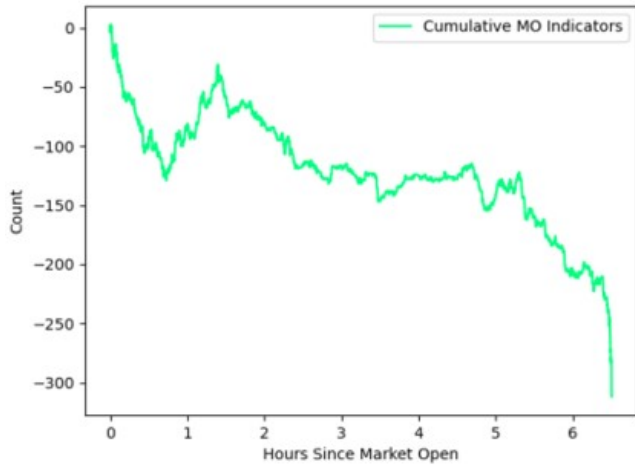


Figure 16: INTC cumulative signed market-order flow $\sum_{j=0}^k \Phi_j$.

7.2 Autocorrelation and PACF Plots

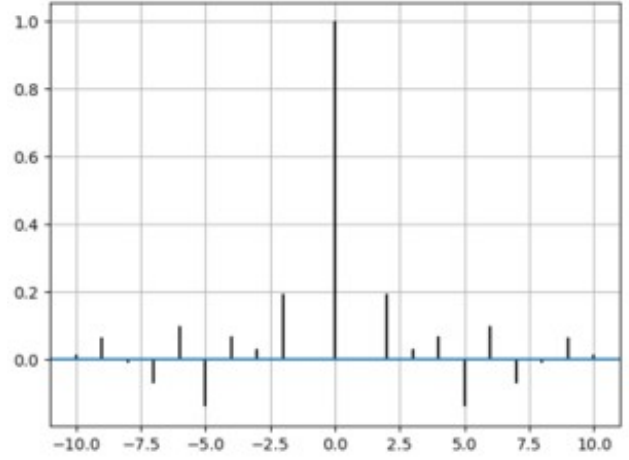


Figure 18: Sample ACF of signed market-order flow Φ_k .

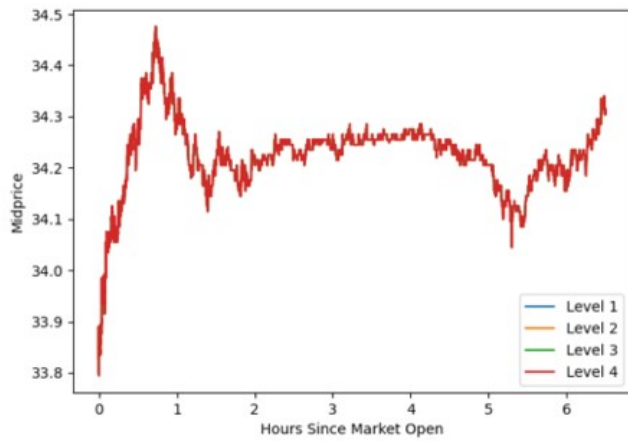


Figure 17: INTC level-1 midprice $M(t)$.

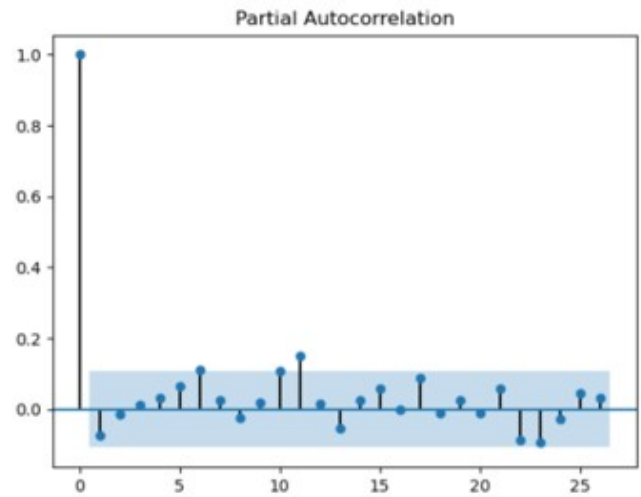


Figure 19: Sample PACF of ΔP_k . No lag is individually significant below lag 11.

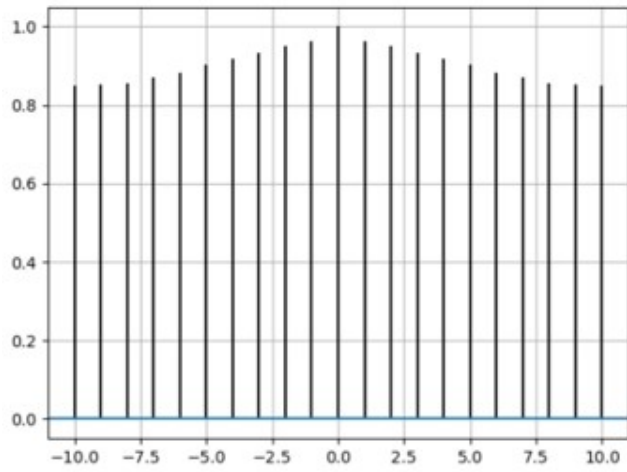


Figure 20: Sample ACF of total volume V_k .

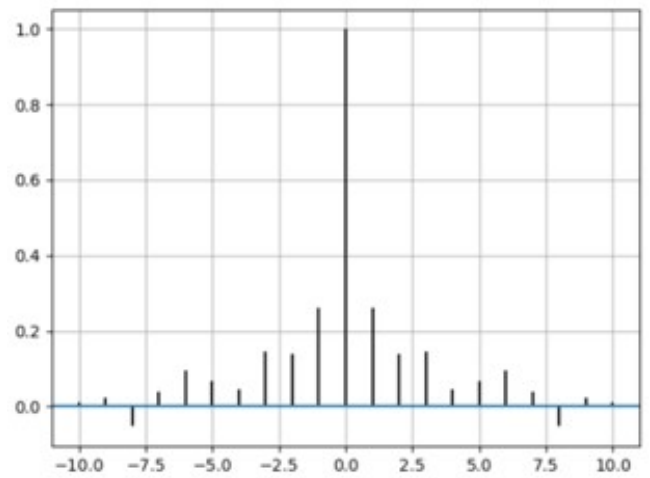


Figure 22: Sample ACF of Φ_k (market-order indicator times volume).

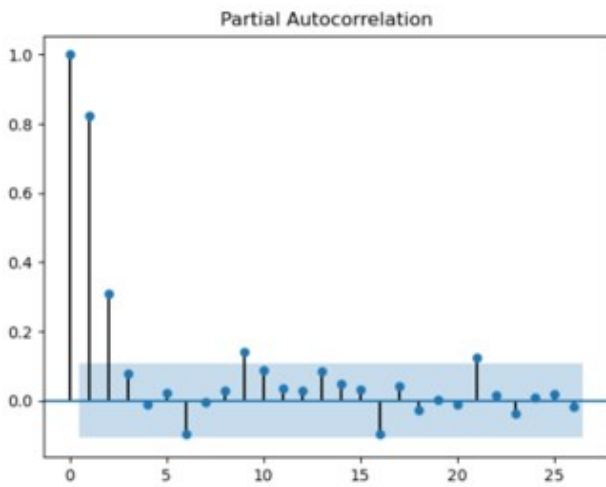


Figure 21: Sample PACF of total volume V_k .

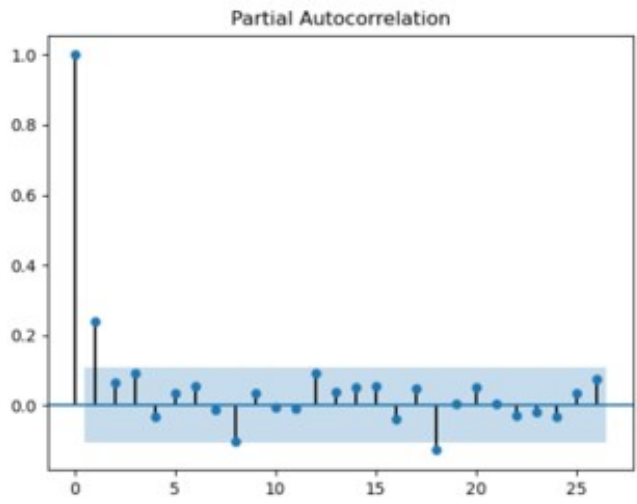


Figure 23: Sample PACF of Φ_k .

7.3 AR Model Summaries

AutoReg Model Results						
=====						
Dep. Variable:	y	No. Observations:	335			
Model:	AutoReg(1)	Log Likelihood	837.026			
Method:	Conditional MLE	S.D. of innovations	0.020			
Date:	Tue, 04 May 2021	AIC	-7.838			
Time:	10:57:49	BIC	-7.815			
Sample:	1	HQIC	-7.829			
	335					
=====						
	coef	std err	z	P> z	[0.025	0.975]

y.L1	-0.0677	0.055	-1.240	0.215	-0.175	0.039
	Roots					
=====						
	Real	Imaginary	Modulus	Frequency		

AR.1	-14.7768	+0.0000j	14.7768	0.5000		

Figure 24: AR(1) MLE model summary for ΔP_k .

AutoReg Model Results						
=====						
Dep. Variable:	y	No. Observations:	335			
Model:	AutoReg(2)	Log Likelihood	834.063			
Method:	Conditional MLE	S.D. of innovations	0.020			
Date:	Tue, 04 May 2021	AIC	-7.829			
Time:	10:55:56	BIC	-7.795			
Sample:	2	HQIC	-7.816			
	335					
=====						
	coef	std err	z	P> z	[0.025	0.975]

y.L1	-0.0682	0.055	-1.245	0.213	-0.176	0.039
y.L2	-0.0079	0.055	-0.144	0.886	-0.115	0.100
	Roots					
=====						
	Real	Imaginary	Modulus	Frequency		

AR.1	-4.3357	-10.4085j	11.2755	0.3128		
AR.2	-4.3357	+10.4085j	11.2755	0.3128		

Figure 25: AR(2) MLE model summary for ΔP_k .

AutoReg Model Results						
=====						
Dep. Variable:	y	No. Observations:	335			
Model:	AutoReg(3)	Log Likelihood	852.468			
Method:	Conditional MLE	S.D. of innovations	0.019			
Date:	Tue, 04 May 2021	AIC	-7.949			
Time:	10:29:42	BIC	-7.903			
Sample:	3	HQIC	-7.931			
	335					
=====						
	coef	std err	z	P> z	[0.025	0.975]

y.L1	-0.0729	0.051	-1.417	0.157	-0.174	0.028
y.L2	-0.0071	0.052	-0.137	0.891	-0.108	0.094
y.L3	0.0162	0.052	0.314	0.753	-0.085	0.117
Roots						
=====						
	Real	Imaginary	Modulus	Frequency		

AR.1	-2.0278	-3.1005j	3.7047	-0.3422		
AR.2	-2.0278	+3.1005j	3.7047	0.3422		
AR.3	4.4925	-0.0000j	4.4925	-0.0000		

Figure 26: AR(3) MLE model summary for ΔP_k .

7.4 OLS Model Summaries

OLS Regression Results					
=====					
Dep. Variable:	y	R-squared (uncentered):	0.006		
Model:	OLS	Adj. R-squared (uncentered):	0.000		
Method:	Least Squares	F-statistic:	1.058		
Date:	Tue, 04 May 2021	Prob (F-statistic):	0.348		
Time:	16:05:26	Log-likelihood:	840.32		
No. Observations:	335	AIC:	-1677.		
Df Residuals:	333	BIC:	-1669.		
Df Model:	2				
Covariance Type:	nonrobust				
=====					
	coef	std err	t	P> t	[0.025 0.975]

x1	-3.373e-10	3.57e-10	-0.946	0.345	-1.04e-09 3.64e-10
x2	2.369e-07	2.17e-07	1.093	0.275	-1.89e-07 6.63e-07
=====					
Omnibus:	111.835	Durbin-Watson:	2.070		
Prob(Omnibus):	0.000	Jarque-Bera (JB):	561.982		
Skew:	1.310	Prob(JB):	9.27e-123		
Kurtosis:	8.779	Cond. No.	608.		
=====					

Figure 27: OLS summary for specification (M1): $\Delta P_k = \alpha_1 V_k + \alpha_2 \Phi_k + \varepsilon_k$.

OLS Regression Results					
=====					
Dep. Variable:	y	R-squared (uncentered):	0.004		
Model:	OLS	Adj. R-squared (uncentered):	0.001		
Method:	Least Squares	F-statistic:	1.221		
Date:	Tue, 04 May 2021	Prob (F-statistic):	0.270		
Time:	16:09:46	Log-Likelihood:	839.88		
No. Observations:	335	AIC:	-1678.		
Df Residuals:	334	BIC:	-1674.		
Df Model:	1				
Covariance Type:	nonrobust				
=====					
	coef	std err	t	P> t	[0.025 0.975]

x1	2.394e-07	2.17e-07	1.105	0.270	-1.87e-07 6.66e-07
=====					
Omnibus:	118.325	Durbin-Watson:	2.069		
Prob(Omnibus):	0.000	Jarque-Bera (JB):	622.994		
Skew:	1.381	Prob(JB):	5.23e-136		
Kurtosis:	9.083	Cond. No.	1.00		
=====					

Figure 28: OLS summary for specification (M2): $\Delta P_k = \alpha_1 \Phi_k + \varepsilon_k$.

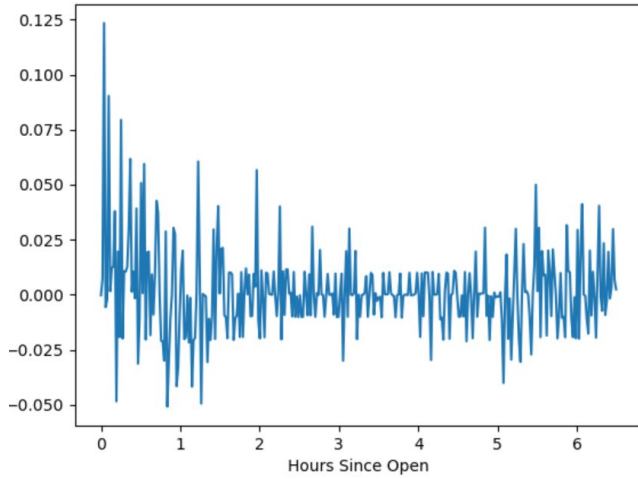


Figure 29: Residual plot for OLS specification (M2).

7.5 VAR(p) Model Summaries

Summary of Regression Results

=====

Model: VAR

Method: OLS

Date: Tue, 04, May, 2021

Time: 15:54:14

No. of Equations: 2.00000 BIC: -2.88803

Nobs: 139999. HQIC: -2.88832

Log likelihood: -195104. FPE: 0.0556624

AIC: -2.88845 Det(Omega_mle): 0.0556601

Results for equation y1

=====

coefficient std. error t-stat prob

const 0.000010 0.000005 1.815 0.070

L1.y1 -0.011108 0.002672 -4.156 0.000

L1.y2 -0.000000 0.000000 -1.911 0.056

=====

Results for equation y2

=====

coefficient std. error t-stat prob

const 81.443400 3.247135 25.082 0.000

L1.y1 -14588.369613 1638.781657 -8.902 0.000

L1.y2 0.990084 0.000376 2636.642 0.000

=====

Correlation matrix of residuals

y1 y2

y1 1.000000 0.054623

y2 0.054623 1.000000

Figure 31: Regime I VAR(1) summary with $\tilde{V}_k$ as second endogenous variable (mid-session).

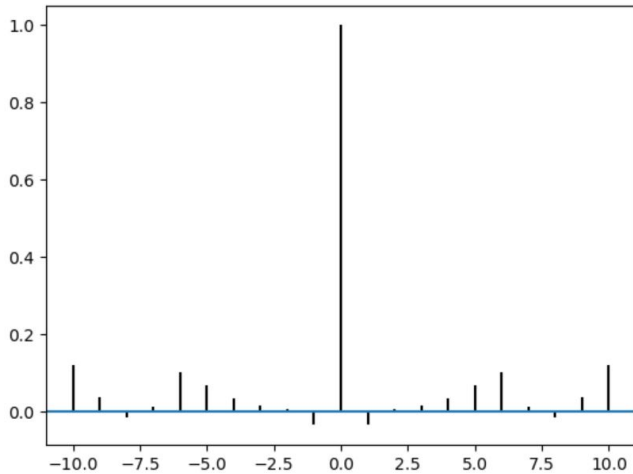


Figure 30: Residual ACF for OLS specification (M2).

=====VAR

Model: VAR

Method: OLS

Date: Tue, 04, May, 2021

Time: 15:42:38

No. of Equations: 2.00000 BIC: -2.49760

Nobs: 49999.0 HQIC: -2.49833

Log likelihood: -79419.7 FPE: 0.0821950

AIC: -2.49866 Det(Omega_mle): 0.0821851

Results for equation y1

=====

	coefficient	std. error	t-stat	prob
const	0.000006	0.000006	0.940	0.347
L1.y1	-0.044426	0.004467	-9.945	0.000
L1.y2	-0.000000	0.000000	-3.632	0.000

=====

Results for equation y2

=====

	coefficient	std. error	t-stat	prob
const	-0.393938	0.912495	-0.432	0.666
L1.y1	-402.874218	648.726187	-0.621	0.535
L1.y2	0.023290	0.004471	5.209	0.000

=====

Figure 32: Regime II VAR(1) summary with Φ_k as second endogenous variable (market open).

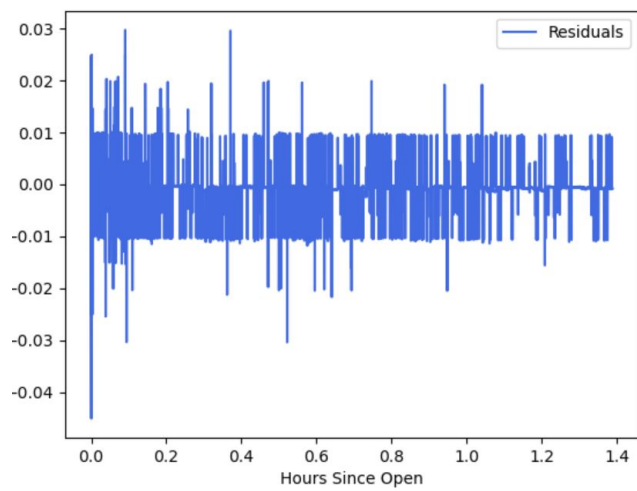


Figure 33: Regime II VAR(1) residuals (Φ_k specification).